

Comparing Chunking Strategies for Retrieval-Augmented Question Answering

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Abstract

We study how document chunking affects retrieval-augmented question answering in a controlled pipeline where the embedding model, retriever, and generator are held fixed. Four chunking families are compared: fixed-size token windows, recursive chunking, sentence-based chunking, and semantic chunking. Using Mistral-7B-Instruct-v0.3 served through vLLM on SCC, we find that recursive_256 is the strongest system on both a 60-question SQuAD v2 subset (60.6 F1) and a 30-question HotpotQA subset (42.6 F1). A parametric-only baseline is substantially weaker, confirming retrieval sensitivity. A fine-grained error analysis of all EM=0 predictions further shows that 65.7% of SQuAD failures and 72.7% of HotpotQA failures are fixable format or refusal errors rather than genuine retrieval or reasoning errors, pointing to prompt-level improvements as the primary path to higher scores.¹

1 Introduction

Retrieval-augmented generation (RAG) answers questions by retrieving external evidence and passing it to a language model (Lewis et al., 2020). In practice, retrieval operates over chunks rather than full documents, making chunking a central design choice: it determines what evidence stays together, how likely the retriever is to recover it, and whether the generator receives coherent support or fragmented context.

This project asks: *which chunking strategy maximizes end-to-end QA accuracy in a fixed RAG pipeline, and does the answer differ between local factoid questions and multi-hop questions?*

Formally, let a corpus be D , a question be q , and a chunking function be $c : D \rightarrow \{d_1, \dots, d_n\}$.

After chunking, each chunk is embedded and indexed. At inference time, the system retrieves the top- K chunks for q , concatenates them into a prompt, and produces an answer $\hat{a}$. The optimization target is end-to-end QA quality:

$$c^* = \arg \max_c \mathbb{E}_{(q,a)} [F_1(\hat{a}, a)].$$

Retrieval quality and answer quality do not always move together: a system may retrieve locally relevant text yet still fail if the evidence is incomplete or badly fragmented. We therefore evaluate chunking with both end-to-end QA metrics and a post-hoc error analysis that attributes each failure to retrieval, generation format, or genuine model error. Figure 1 summarises the full pipeline.

2 Related Work

RAG conditions a language model on retrieved passages rather than relying on parametric memory alone (Lewis et al., 2020). Dense Passage Retrieval (Karpukhin et al., 2020) established bi-encoder dense retrieval as a strong backbone for open-domain QA and serves as our fixed retriever. We evaluate on SQuAD 2.0 (Rajpurkar et al., 2018), which tests single-hop answer extraction, and HotpotQA (Yang et al., 2018), which requires composing evidence across multiple documents, making it a natural pair for studying whether chunking effects depend on question complexity.

Chunking has recently attracted attention as a first-class experimental variable. Smith and Troynikov (2024) find that chunk size and overlap substantially affect retrieval precision and recall, with no single strategy dominating across domains. NVIDIA’s end-to-end study (NVIDIA, 2024) shows that retrieval metrics alone can be misleading and that chunking should be evaluated on full pipeline QA performance. Qu et al. (2024) question whether semantic chunking consistently

¹Code and configs: <https://github.com/kuromilkow/chunkrag-course-project>

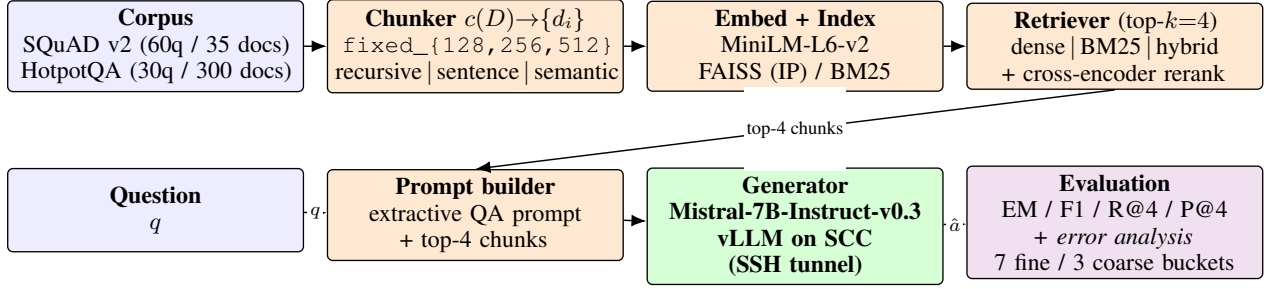


Figure 1: System architecture. The chunker is the only variable; embedding model, index, retriever (top- $k=4$), prompt template, generator (Mistral-7B-Instruct-v0.3 served via vLLM on SCC over an SSH tunnel), and evaluation are held fixed across all runs. Recall@4/Precision@4 are computed at the document level; the error analyser assigns each EM=0 prediction to one of seven fine buckets that collapse into three coarse buckets (retrieval / format-or-refusal / model).

justifies its computational cost, finding inconsistent gains across benchmarks. We build on these findings by comparing the four major chunking families in a single controlled pipeline and adding a post-hoc error analysis to separate retrieval failures from generation failures.

3 Methods

For each chunking strategy, the pipeline chunks each document, embeds every chunk with the same encoder, builds a FAISS dense index, retrieves the top 4 chunks for each question, and passes them to the same generator. The chunking function is the only variable.

3.1 Chunking Strategies

We implement four families:

- **Fixed token chunking**(fixed_128/256/512, ~15% overlap): fixed-length windows ignoring sentence or paragraph boundaries. Overlap mitigates boundary loss but coherent passages can still be split awkwardly.
- **Recursive chunking**(recursive_256, LangChain): an ordered cascade of separators (paragraph $\rightarrow$ line break $\rightarrow$ sentence punctuation $\rightarrow$ character) that keeps larger discourse units intact whenever they fit the token budget.
- **Sentence chunking**(sentence_256, spaCy): segments first into sentences and greedily packs consecutive sentences up to the budget, respecting sentence boundaries but still able to split related evidence.

- **Semantic chunking**(semantic_256): splits when inter-sentence cosine similarity drops below 0.72 (with a minimum-length guard), so boundaries follow embedding-detected topical shifts rather than token counts.

3.2 Models and Configuration

The pipeline uses all-MiniLM-L6-v2 for embedding, FAISS inner-product search over normalized embeddings for retrieval, and Mistral-7B-Instruct-v0.3 served through vLLM on SCC as the generator. Prompting consists of a short extractive QA instruction plus a lightweight answer-compression step for verbose outputs. Generation is limited to 1024 input tokens and 48 new tokens. The broader codebase also supports BM25, hybrid retrieval, and reranking, but the analysis keeps the ablation focused on chunking. All hyperparameters are fixed in JSON configs and reused across runs.

3.3 Baseline

Our baseline is **parametric-only QA**: the generator answers without retrieved context. This tests whether retrieval is necessary and provides a lower bound.

3.4 Error Analysis

To attribute EM=0 failures, we assign each incorrect prediction to exactly one of seven categories in priority order (earlier categories take precedence when multiple could apply):

- *retrieval failure*: gold document absent from the top-4 chunks;
- *false refusal*: model outputs “unanswerable” despite adequate retrieval;

- *verbose extraction*: gold tokens are a strict subset of predicted tokens (right answer, extra words added);
- *terse extraction*: predicted tokens are a strict subset of gold tokens (correct but incomplete span);
- *close paraphrase*: $F1 \geq 0.6$, neither is a token-set subset of the other;
- *partial answer*: $0 < F1 < 0.6$; and
- *wrong answer*: $F1 = 0$.

For summary, categories 2–5 are grouped as *format or refusal*, all fixable by prompt or post-processing changes without touching the retriever or model weights.

4 Experiments

4.1 Datasets

- **SQuAD v2**: 60 answerable validation questions sampled from a 500-example pool. Paragraphs sharing a title are concatenated into 35 article-level pseudo-documents to make chunking non-trivial.
- **HotpotQA distractor**: 30 validation questions with 300 documents built from distractor and supporting titles.

4.2 Evaluation

We report Exact Match (EM), token-level F1, Recall@4, and Precision@4 as quality metrics, and include average chunk size and chunk count as corpus statistics describing each chunker. Recall@4 is defined as the fraction of gold supporting documents that appear in the top-4 retrieved chunks. For SQuAD, each question has exactly one gold document, so Recall@4 is binary per question: 1 if any retrieved chunk came from that document, 0 otherwise. For HotpotQA, each question has two gold documents, so Recall@4 can take values 0, 0.5, or 1.0, making it a more informative measure of evidence coverage for multi-hop questions.

5 Results

All results below are from single runs on the SQuAD v2 subset ($n = 60$) and HotpotQA subset ($n = 30$).

The parametric-only baseline is weak on both datasets, confirming retrieval sensitivity. On

SQuAD, `recursive_256` leads at 60.6 F1 and 41.7 EM, with `semantic_256` competitive at 56.9 F1 and tied for the best Recall@4 (95.0). `fixed_128` achieves the highest retrieval precision (77.1) but lower F1, suggesting that very small chunks recover evidence but may fragment it. Larger fixed windows (`fixed_512`) degrade F1 (53.7 vs. 56.7 for `fixed_256`), consistent with the hypothesis that broad chunks dilute relevant evidence with distractors; EM edges up slightly to 40.0, suggesting that length constraints hurt token-level overlap more than strict span matching.

On HotpotQA, `recursive_256` is best at 42.6 F1 and 26.7 EM, with `semantic_256` essentially tied on F1 at 42.4. Doc-level Recall@4 tops out at 76.7 rather than saturating, suggesting that evidence composition rather than chunk boundary placement is the main bottleneck once retrieval is only partially successful. A further sign that retrieval coverage and answer quality are decoupled: on SQuAD, `fixed_128` and `semantic_256` share the best Recall@4 (95.0) yet land 6.2 F1 points apart, and on HotpotQA four of six chunkers sit at Recall@4=76.7 with F1 spread from 34.3 to 42.6.

5.1 Error Analysis

Table 3 shows the coarse failure breakdown for `recursive_256`. The majority of EM=0 cases are fixable format or refusal errors: 65.7% on SQuAD and 72.7% on HotpotQA. Retrieval accounts for only 14.3% of SQuAD failures and zero HotpotQA failures. The remaining 20–27% are genuine model errors where supporting evidence was present but the answer is substantially wrong. Figure 2 extends this view to all six chunkers and confirms the pattern: format-or-refusal errors dominate the EM=0 bucket for every chunker on both datasets, retrieval failure is at most 20% on SQuAD and zero on HotpotQA, and genuine model errors stay in the 20–40% range.

The fine-grained breakdown (Table 4) reveals that the dominant failure mode differs by dataset. On SQuAD, verbose extraction accounts for 40.0% of EM=0 cases: the gold span is present inside the prediction but flanked by extra words. On HotpotQA, false refusals are the largest single category at 40.9%: Mistral outputs *unanswerable* for nearly four in ten of its mistakes. All four SQuAD false refusals had Recall@4=1.0, and four of nine HotpotQA false refusals likewise

System	EM	F1	Recall@4	Precision@4	Avg chunk tokens	# chunks
parametric_only	13.3	29.2	0.0	0.0	–	–
fixed_128	38.3	50.7	95.0	77.1	125.3	631
fixed_256	38.3	56.7	88.3	73.8	244.0	322
fixed_512	40.0	53.7	90.0	63.7	472.2	164
recursive_256	41.7	60.6	91.7	72.5	175.8	389
sentence_256	35.0	51.0	86.7	72.1	223.9	302
semantic_256	41.7	56.9	95.0	74.6	144.8	467

Table 1: Results on SQuAD v2. `recursive_256` is the strongest end-to-end setting.

System	EM	F1	Recall@4	Precision@4	Avg chunk tokens	# chunks
parametric_only	10.0	27.7	0.0	0.0	–	–
fixed_128	20.0	36.5	75.0	42.5	89.3	421
fixed_256	20.0	34.3	75.0	40.8	114.2	313
fixed_512	20.0	35.7	76.7	41.7	117.4	301
recursive_256	26.7	42.6	76.7	42.5	113.6	313
sentence_256	23.3	37.0	76.7	41.7	112.7	313
semantic_256	23.3	42.4	76.7	42.5	97.2	363

Table 2: Results on HotpotQA. Doc-level Recall@4 shows that recovering all supporting documents remains incomplete.

Failure type	SQuAD		HotpotQA	
	<i>n</i>	%	<i>n</i>	%
Retrieval failure	5	14.3	0	0.0
Format or refusal (fixable)	23	65.7	16	72.7
Model error	7	20.0	6	27.3
Total EM=0	35	100.0	22	100.0

Table 3: Coarse failure breakdown for `recursive_256`. “Format or refusal” groups verbose extraction, terse extraction, close paraphrases, and false refusals, all fixable without changing the retriever.

had full supporting-document coverage, confirming that the evidence was in the prompt each time the model refused. Annotated examples of each failure type are provided in Appendix A.

Failure type	SQuAD		HotpotQA	
	<i>n</i>	%	<i>n</i>	%
Retrieval failure	5	14.3	0	0.0
False “unanswerable”	4	11.4	9	40.9
Verbose extraction	14	40.0	5	22.7
Terse extraction	2	5.7	2	9.1
Close paraphrase	3	8.6	0	0.0
Partial answer	4	11.4	1	4.5
Wrong answer	3	8.6	5	22.7

Table 4: Fine-grained breakdown of EM=0 predictions for `recursive_256`.

6 Discussion

Two findings stand out from this study. First, chunking strategy has a measurable, dataset-dependent effect on end-to-end QA quality. Second, the error analysis reveals that the headline EM scores understate true system quality: the majority of failures are prompt artifacts addressable without changing the retriever or model.

Chunking matters, and the best strategy depends on question type. On SQuAD, chunkers preserving moderately large coherent spans (recursive, semantic) outperform both very small fixed windows and sentence-level segmentation: the F1 drop at `fixed_512` (despite a slight EM uptick) shows broad chunks dilute token-level evidence with distractors, while `fixed_128`’s strong retrieval precision but weaker F1 indicates very small chunks fragment evidence that should stay together. On HotpotQA, EM is flatter but F1 still varies, and fine-grained fixed chunks fail to help multi-hop questions despite high individual-fact recall, suggesting those questions need chunks large enough to preserve the local structure linking related facts.

Most failures are fixable, and the dominant mode differs by dataset. On SQuAD the largest failure category is verbose extraction (40% of EM=0): the correct span appears inside the prediction surrounded by extra words, a systematic artifact of the prompt’s lack of length constraints. On HotpotQA false refusals dominate (41%): Mis-

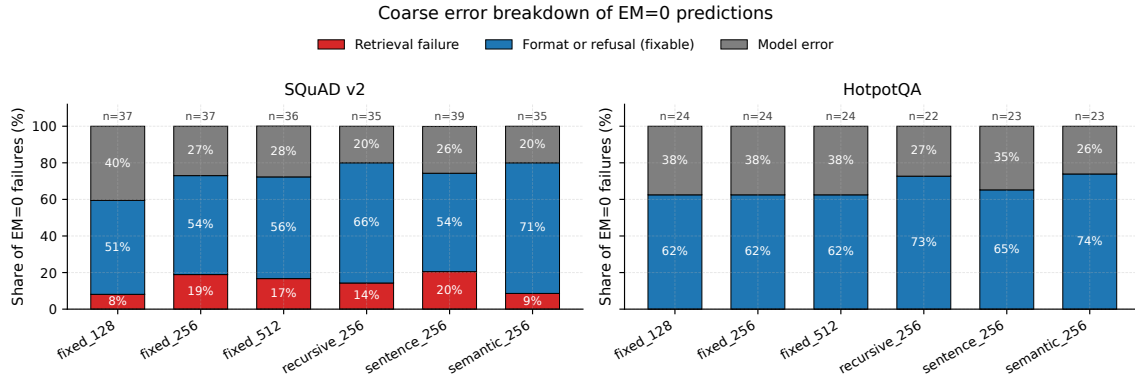


Figure 2: Coarse error breakdown of EM = 0 predictions by chunker. Each bar is normalised to 100%; the absolute number of EM = 0 predictions is annotated above each bar. “Format or refusal” aggregates verbose extraction, terse extraction, close paraphrases, and false refusals (categories fixable without changing the retriever or model). The fixable share never drops below 51% on any chunker–dataset cell.

tral outputs *unanswerable* even when all supporting evidence is in the prompt, plausibly because synthesising facts across passages raises uncertainty and the “say unanswerable if unsure” instruction provides an easy exit. Both modes are addressable by targeted prompt changes alone.

7 Conclusion

Chunking is a non-trivial experimental variable: across two QA benchmarks with different question structures, `recursive_256` is the most consistently strong chunker, and the parametric-only baseline falls well below every retrieval-based system. The dataset-level difference, where coherence-preserving chunkers dominate on single-hop SQuAD while the gap narrows on multi-hop HotpotQA, supports the hypothesis that the optimal chunking policy depends on question complexity.

The error analysis qualifies these conclusions in an important way. The majority of EM = 0 failures are fixable format or refusal errors rather than genuine retrieval or reasoning mistakes, and the dominant failure mode differs by dataset: verbose extraction on SQuAD and false refusals on HotpotQA. These findings suggest that dataset-specific prompt improvements are the highest-leverage next step, and that once prompt issues are resolved, the chunking comparison can be extended cleanly to BM25, hybrid, and reranked retrieval settings already implemented in the codebase.

Progress Since Midway Report

Since the midway report we completed the controlled evaluation across both SQuAD v2 and HotpotQA, added the parametric-only baseline, finalised all six chunking configurations, and extended the metric set beyond EM/F1 to document-level Recall@4 and Precision@4. We also incorporated the midway feedback by tightening the experimental framing so that the chunking function is the only controlled variable, and by adding the post-hoc error analysis that separates retrieval failures from generation-format and model-reasoning failures. This shifted the project’s final reading: rather than treating low EM as primarily a retrieval problem, we show that a majority of failures are prompt- or formatting-related and therefore fixable without changing the retriever or the generator weights.

Author Contributions

Assylkhan Geniyat led the engineering effort and authored the bulk of the codebase: the chunking, retrieval, and generation modules in `src/chunkrag`; the BM25, hybrid, and cross-encoder rerank baselines; the SCC `vLLM` Mistral deployment together with the SSH-tunnel toolkit; the OpenWebUI integration and the Streamlit RAG playground; the unit-test suite; the iterative prompt tuning that closed most of the early-stage gap on SQuAD; the `Makefile` one-command build; and the ACL report scaffold. **Jaime Bernal** designed and implemented the post-hoc error-analysis pipeline (the 7-fine / 3-coarse failure taxonomy and the EDA report that under-

pins the error-analysis results), authored the Discussion, Limitations, and Remaining Work sections of the midway report, and shaped the failure-mode framing carried into the final manuscript. **Batyrkhan Baimukhanov** contributed the Mistral and Qwen Colab experiment notebooks, the chunking-strategy descriptions, and the failure-analysis write-up integrated into the final report. **Pablo Bello** built the publication-figure pipeline used for Figures 2–4 and integrated them into the manuscript, and reviewed the architecture diagram and final-report layout for technical and editorial consistency. All authors discussed methodology and reviewed the final manuscript.

8 Limitations

Strict metrics. EM/F1 penalise surface-form mismatches even when semantic content is correct; the error analysis quantifies this (40% of SQuAD EM = 0 cases are verbose extractions with the gold span inside the prediction). **Uniform prompting.** A single extractive QA prompt is used for both datasets; HotpotQA would benefit from a dataset-specific prompt that softens the unanswerable instruction and surfaces document titles. **Limited post-processing.** The answer normaliser does not cover all observed mismatch patterns (determiners, parentheticals, units). **Small evaluation subsets.** 60 and 30 questions reveal broad trends but the ranking among chunkers could shift with larger samples.

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A Failure Examples

Four representative annotated failures from recursive_256, one per category in the coarse breakdown of Table 3: *retrieval failure*, *false refusal* (the dominant fixable mode on HotpotQA), *verbose extraction* (the dominant fixable mode on SQuAD), and *wrong answer* (genuine model error). Predictions are verbatim.

Retrieval failure (SQuAD) *Q*: What entity owns V/Line? *Gold*: Victorian Government *Pred*: unanswerable *F1*: 0.00, *R@4*: 0.00. The retriever missed the gold article entirely; unrecoverable by prompt changes alone.

False refusal (HotpotQA) *Q*: Robert Earl Holding owned an oil company originally founded by whom? *Gold*: Harry F. Sinclair *Pred*: unanswerable *F1*: 0.00, *R@4*: 1.00. All four supporting documents were in the prompt; the model declined despite having the evidence—fixable by softening the unanswerable instruction.

Verbose extraction (SQuAD) *Q*: What is thought to have happened to the *Y. pestis* that caused the Black Death? *Gold*: may no longer exist *Pred*: *Y. pestis* may no longer exist *F1*: 0.80, *R@4*: 1.00. Gold span is present verbatim inside the prediction; EM fails, F1 is penalised by extra tokens. Suppressing subject echoing recovers this case.

Wrong answer (HotpotQA) *Q*: Which mountain is taller, Gasherbrum II or Langtang Ri? *Gold*: Gasherbrum II *Pred*: taller than Langtang Ri *F1*: 0.00, *R@4*: 1.00. Both articles retrieved; the model named the wrong mountain. Genuine reasoning failure not attributable to format or retrieval.